

11102

Introduction to Computing

Spring 2026

Agenda

- Get to know **me**.
- Get to know **you**.
- Get to know the **course**.

Warmup Challenge # 1

You have 10,000 black boxes, all of the same weight, except **one**!!

You want to use a **scale** to find this **heavy** box. What should you do?

Assume the scale can hold any number of boxes.



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Solution # 1

Compare every pair until two of different weight are found.



Too Slow!!

Warmup Challenge # 1

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You want to use a **scale** to find this **heavy** box. What should you do?

Assume the scale can hold any number of boxes.

Solution # 2

Split the boxes into **2 piles of 5,000** boxes each. Use the scale to compare them.
The heavier pile definitely has the heavier box! Get rid of the other pile!

Split the 5,000 pile into two piles of size **2,500**.
Compare and get rid of the lighter pile.

Repeat until we are left with only **2 boxes**.

Each use of the scale **eliminates half** of the boxes!



Very Quick!

Note

An **important** challenge in computing is:

How do we automate a task such that it is achieved in as **little time** as possible?



You will get
a *taste* of this
in this course!

Warmup Challenge # 2

You want to know the **average age** of the people in a room, but no one in the room is willing to reveal their age to anyone.

What should you do?



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Solution # 1

Ask everyone to write their age and put it in a box.
Shuffle the box and then open it and compute the average.

Does this guarantee privacy? 🤔

No. You can say something like: *I know none of you is below 40!*

This will make some quite angry!

Warmup Challenge # 2

You want to know the **average age** of the people in a room, but no one in the room is willing to reveal their age to anyone.

What should you do?



Solution # 2

- Write a **random number** on a piece of paper.
- Give it to the first person and ask them to **add their age**, write the sum on another paper, and pass it to the next person (the other paper must be destroyed).
- The next person does the same, and so on.
- When everyone is done, subtract the random number from the total.



Bingo!

Note

An **important** challenge in computing is:

How do we perform tasks while
maintaining **privacy**?



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in this course!

Warmup Challenge # 3

I made a **super-intelligent paper**.
Do you believe me?

If not, do you want to challenge it?



Note

An **important** challenge in computing is:

How do we make **intelligent** machines?



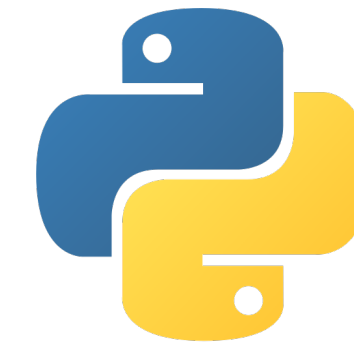
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What is this course about?

- It is an introduction to the field of **computer science**!

- We will do that through programming in **Python**.

A very popular and quite simple programming language!



- We will get a taste of:

- Algorithms.
- Software Engineering.
- Computer graphics and image processing.
- Computer security.
- Artificial intelligence.

first 7 weeks

6 weeks
one week per
application
domain

Assessment	Weight	Description
Exercises	5%	11 problemsets due at the end of every week. You are allowed to drop the lowest 2.
Self Assessments	5%	11 quizzes for self assessment purposes done at home and due at the end of every week. You are allowed to drop the lowest 2.
Quizzes	12%	Two quizzes done in class, each worth 6%.
Assignments	8%	Three practical assignments. Details to be announced later.
Midterm Exam	30%	The exact date and time will be set by the university.
Final Exam	40%	The exact date and time will be set by the university.
Total	100%	

Learning Material



<https://ialbluwi.github.io/11102-s26>

Course information, policies, textbook, slides, notes, etc.



Email Invitation will be sent soon!

Class discussion, announcements, and exercises.



Already enrolled.

Midterm and Final Exams.

Section-specific material.

Schedule

Su/Tue/Thu

Mo/We

Week 1 (Introduction)

Feb 15: Course Introduction

POLICY QUIZ

[\[Slides\]](#) [\[Heavy Box Game\]](#)

Feb 17: Introduction to Python

[P4E.1](#)

Install: [python](#), [vscode](#), [python for vscode](#), [GitHub Copilot](#)

Feb 19: Variables and Expressions

[PE4.2](#)

HW:

EXERCISES

SELF-ASSESSMENT

Due over the weekend!

Are you ready?!



Yes!!!

